

Grad Cafe SQL Analysis

Nipul Jayasekera (njayase1) EN.605.256 Modern Software Concepts in Python - Module 3

Eleven questions answered against a PostgreSQL table of 50,005 Grad Cafe admissions results. Every query below was executed through psycopg by query_data.py, and this document is generated from that run, so the result printed beside each query is the one that query produced rather than a figure copied across by hand. Counts are whole numbers; percentages and averages are given to two decimal places.

Question 1

How many entries in the database are from applicants who applied for Fall 2026?

RESULT

Fall 2026 applicant count: 33,208

SQL

```
SELECT COUNT(*) AS fall_2026_entries
FROM applicants
WHERE lower(trim(term)) = 'fall 2026';
```

WHAT IT DOES

Counts every row whose term is Fall 2026. COUNT(*) is used rather than COUNT(term) because the filter has already guaranteed the column is non-NULL, and lower(trim(...)) makes the match insensitive to capitalisation and stray whitespace.

Question 2

Among entries that provide a nationality classification, what percentage are international students?

RESULT

Percent international: 47.89%

SQL

```
SELECT ROUND(
    100.0 * COUNT(*) FILTER (WHERE lower(trim(us_or_international)) = 'international')
    / NULLIF(COUNT(*), 0),
    2
) AS percent_international
FROM applicants
WHERE us_or_international IS NOT NULL
AND trim(us_or_international) <> '';
```

WHAT IT DOES

The WHERE clause sets the denominator to entries carrying a usable classification, so missing values are excluded rather than counted as domestic. FILTER supplies the numerator from that same scan. 'American' and 'Other' therefore sit in the denominator but not the numerator, which is what the question asks. NULLIF guards against division by zero on an empty table.

Question 3

What are the average GPA, GRE Quantitative, GRE Verbal and GRE Analytical Writing scores of applicants who provide each metric?

RESULT

Average GPA: 3.77

Average GRE Quantitative: 261.45
 Average GRE Verbal: 161.14
 Average GRE Analytical Writing: 9.17

Metric	Reported	Out of range	Average (all)	Average (in range)	Distortion
GPA (0-4.0)	29,483	231	3.77	3.75	+0.02
GRE Analytical Writing (0-6)	2,938	153	9.17	4.33	+4.84
GRE Verbal (130-170)	3,139	15	161.14	160.37	+0.77
GRE Quantitative (130-170)	3,769	2,316	261.45	165.74	+95.71

SUPPORTING ANALYSIS

- How much of each average is an artefact of impossible values:
- 2,264 of the 2,316 impossible GRE Quantitative values (97.75%) fall in 260-340, the official combined Verbal+Quantitative range
- Subtracting the verbal score from the 1,675 of those that report one leaves a mean of 165.26 — back inside the valid 130-170 band

SQL

```

SELECT ROUND(AVG(gpa)::numeric, 2) AS avg_gpa,
       ROUND(AVG(gre)::numeric, 2) AS avg_gre_quant,
       ROUND(AVG(gre_v)::numeric, 2) AS avg_gre_verbal,
       ROUND(AVG(gre_aw)::numeric, 2) AS avg_gre_aw
FROM applicants;

-- supporting: what those averages are worth
SELECT 'GPA (0-4.0)' AS metric,
       COUNT(gpa) AS reported,
       COUNT(*) FILTER (WHERE gpa > 4.0) AS out_of_range,
       ROUND(AVG(gpa)::numeric, 2) AS average_all,
       ROUND(AVG(gpa) FILTER (WHERE gpa <= 4.0)::numeric, 2) AS average_in_range
FROM applicants
UNION ALL
SELECT 'GRE Quantitative (130-170)',
       COUNT(gre),
       COUNT(*) FILTER (WHERE gre < 130 OR gre > 170),
       ROUND(AVG(gre)::numeric, 2),
       ROUND(AVG(gre) FILTER (WHERE gre BETWEEN 130 AND 170)::numeric, 2)
FROM applicants
UNION ALL
SELECT 'GRE Verbal (130-170)',
       COUNT(gre_v),
       COUNT(*) FILTER (WHERE gre_v < 130 OR gre_v > 170),
       ROUND(AVG(gre_v)::numeric, 2),
       ROUND(AVG(gre_v) FILTER (WHERE gre_v BETWEEN 130 AND 170)::numeric, 2)
FROM applicants
UNION ALL
SELECT 'GRE Analytical Writing (0-6)',
       COUNT(gre_aw),
       COUNT(*) FILTER (WHERE gre_aw < 0 OR gre_aw > 6),
       ROUND(AVG(gre_aw)::numeric, 2),
       ROUND(AVG(gre_aw) FILTER (WHERE gre_aw BETWEEN 0 AND 6)::numeric, 2)
FROM applicants;

-- supporting: what the impossible values actually are
SELECT COUNT(*) FILTER (WHERE gre IS NOT NULL AND (gre < 130 OR gre > 170)) AS impossible,
```

```

COUNT(*) FILTER (WHERE gre BETWEEN 260 AND 340)                AS in_total_range,
COUNT(*) FILTER (WHERE gre BETWEEN 260 AND 340 AND gre_v IS NOT NULL) AS with_verbal,
ROUND(AVG(gre - gre_v) FILTER (
    WHERE gre BETWEEN 260 AND 340 AND gre_v IS NOT NULL)::numeric, 2)
    AS implied_quant
FROM applicants;

```

WHAT IT DOES

SQL's AVG skips NULLs, so each average is taken over exactly the applicants who reported that one metric. Computing all four in a single statement keeps them independent — an applicant with a GPA but no GRE still contributes to the GPA average, which is what the question requires. The two supporting statements are not part of the answer: the first re-computes each average with the impossible values removed so the distortion is visible, and the second tests what those values are rather than merely counting them. Both of the second one's predictions hold, which is why the caveat below states a diagnosis rather than a guess.

READ BEFORE QUOTING THIS NUMBER

These are the averages of exactly what applicants reported, which is what the question asks for — but two of them are not usable as test scores. The GRE reports Verbal and Quantitative on 130-170 each, a combined Verbal+Quantitative total on 260-340, and Analytical Writing on 0-6. The Quantitative column here is bimodal, and the second mode falls in exactly the 260-340 band: those applicants entered their combined total into the box the site labels only 'GRE'. The two supporting checks above confirm that rather than assume it — almost all of the impossible values land inside the official total range, and subtracting the verbal score the same rows report leaves a mean back inside 130-170. Analytical Writing is inflated the same way by placeholder values of 99.99 on a scale that stops at 6. The table above gives each average restricted to the values its scale permits; [limitations.pdf](#) discusses what follows from it.

Question 4

What is the average GPA of American applicants who applied for Fall 2026?

RESULT

Average GPA American: 3.79

SQL

```

SELECT ROUND(AVG(gpa)::numeric, 2) AS avg_gpa_american_fall_2026
FROM applicants
WHERE lower(trim(term)) = 'fall 2026'
    AND lower(trim(us_or_international)) = 'american'
    AND gpa IS NOT NULL;

```

WHAT IT DOES

Three conditions applied together: the Fall 2026 term, an American classification, and a reported GPA. The explicit 'gpa IS NOT NULL' is redundant because AVG ignores NULLs anyway, but it states the intent of the question in the query itself.

Question 5

What percentage of Fall 2025 entries are acceptances?

RESULT

Fall 2025 acceptance percentage: 40.74%

SQL

```

SELECT ROUND(
    100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%')
    / NULLIF(COUNT(*), 0),
    2

```

```

    ) AS fall_2025_acceptance_percent
FROM applicants
WHERE lower(trim(term)) = 'fall 2025';

```

WHAT IT DOES

The denominator is every Fall 2025 entry, as specified — including rejections, waitlists and interviews — while FILTER counts only those whose cleaned status begins with 'Accept'. One pass over the table produces both halves of the fraction.

Question 6

What is the average GPA of accepted applicants who applied for Fall 2026?

RESULT

Average GPA Acceptance: 3.78

SQL

```

SELECT ROUND(AVG(gpa)::numeric, 2) AS avg_gpa_accepted_fall_2026
FROM applicants
WHERE lower(trim(term)) = 'fall 2026'
    AND status ILIKE 'accept%'
    AND gpa IS NOT NULL;

```

WHAT IT DOES

The same shape as Question 4, with the acceptance status swapped in for the nationality filter. Comparing the two answers against Question 3's overall GPA shows how little the reported GPA separates accepted applicants from the field as a whole.

Question 7

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

RESULT

JHU Masters Computer Science count: 18

SQL

```

SELECT COUNT(*) AS jhu_masters_cs_entries
FROM applicants
WHERE (program ILIKE '%johns hopkins%' OR program ~* '\mjhu\M')
    AND program ILIKE '%computer science%'
    AND (degree ILIKE 'master%'
        OR lower(trim(degree)) IN ('ms', 'm.s.', 'msc', 'm.sc.', 'meng', 'm.eng.'));

```

WHAT IT DOES

Uses the original downloaded fields, as the question requires. 'program' holds '<program>, <university>', so one ILIKE matches the university half and another the department half. 'JHU' is matched with the word boundaries \m and \M rather than '%jhu%', which would also fire inside unrelated words. The degree test accepts the cleaned 'Masters' label plus the common abbreviations, but not MFA, MBA or JD, which are different credentials.

Question 8

How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford or Carnegie Mellon? (original downloaded fields)

RESULT

Original-field count: 30

SQL

```
SELECT COUNT(*) AS cs_phd_acceptances_fall_2026
FROM applicants
WHERE lower(trim(term)) = 'fall 2026'
  AND status ILIKE 'accept%'
  AND degree ~* '^ph\.\?\s*d'
  AND program ILIKE '%computer science%'
  AND (
    program ILIKE '%georgetown%'
    OR program ILIKE '%massachusetts institute of technology%'
    OR program ~* '\mmit\M'
    OR program ILIKE '%stanford%'
    OR program ILIKE '%carnegie mellon%'
    OR program ~* '\mcmu\M'
  );
```

WHAT IT DOES

All five restrictions are applied at once in a single WHERE clause: term, status, degree, department and university. The university test is the bracketed OR group, so it narrows the row set rather than widening it — without the parentheses the OR would defeat every AND above it and return far too many rows.

Question 9

Repeat Question 8 using the LLM-generated program and university fields, and compare the two counts.

RESULT

```
Original-field count: 30
LLM-field count: 30
Difference: +0
```

SQL

```
SELECT COUNT(*) AS cs_phd_acceptances_fall_2026_llm
FROM applicants
WHERE lower(trim(term)) = 'fall 2026'
  AND status ILIKE 'accept%'
  AND degree ~* '^ph\.\?\s*d'
  AND llm_generated_program ILIKE '%computer science%'
  AND (
    llm_generated_university ILIKE '%georgetown%'
    OR llm_generated_university ILIKE '%massachusetts institute of technology%'
    OR llm_generated_university ~* '\mmit\M'
    OR llm_generated_university ILIKE '%stanford%'
    OR llm_generated_university ILIKE '%carnegie mellon%'
    OR llm_generated_university ~* '\mcmu\M'
  );
```

WHAT IT DOES

Identical to Question 8 except that the department and university come from the standardized columns, while term, degree and status still come from the original downloaded fields as instructed.

READ BEFORE QUOTING THIS NUMBER

The two counts are equal here, and not by coincidence: they select the identical set of rows, not merely the same number of them. Grad Cafe stores each result's school as a foreign key into its own controlled vocabulary rather than as applicant free text, so these four well-known universities already arrive with one full, correctly spelled name ('Massachusetts Institute of Technology (MIT)', 'Stanford University'). Standardizing a name that is already canonical cannot change which rows match. The standardizer is not idle overall — it rewrites the university on 4,768 of the 50,000 rows, folding 'University of Wisconsin - Madison' and 'University of

California (UCLA)' onto single canonical spellings — but none of that work falls inside this particular filter. A difference would appear for schools the site stores inconsistently, or for a department name like 'CS' that only the LLM column spells out.

Question 10 —

Of the twenty universities that applicants apply to most, which is hardest to get into, and does a lower acceptance rate come with a stronger GPA among those who are accepted?

RESULT

Hardest of the twenty most-applied-to universities: Yale University at 18.47% (152 of 823 entries), average GPA of those accepted 3.82
Easiest: University of Oxford at 52.12% (307 of 589 entries), average GPA of those accepted 4.00
Acceptance rate across the twenty spans 18.47% to 52.12%, but the average GPA of those accepted spans only 3.71 to 4.00 – selectivity barely shows up in the GPA of who gets in

University	Entries	Acceptances	Acceptance rate	Accepted w/ GPA	Avg GPA (accepted)
Yale University	823	152	18.47%	96	3.82
Massachusetts Institute of Technology	731	135	18.47%	86	3.90
Princeton University	719	147	20.45%	83	3.90
University of Wisconsin–Madison	644	135	20.96%	86	3.81
Brown University	654	142	21.71%	95	3.81
Stanford University	1,104	246	22.28%	178	3.87
Harvard University	786	183	23.28%	124	3.81
Northwestern University	716	176	24.58%	115	3.77
University of California, Berkeley	853	225	26.38%	151	3.81
Cornell University	705	191	27.09%	123	3.74
University of Michigan, Ann Arbor	661	180	27.23%	120	3.85
University of Pennsylvania	854	245	28.69%	157	3.77
University of Washington	926	275	29.70%	172	3.71
University of Chicago	764	250	32.72%	169	3.81
University of Texas at Austin	602	212	35.22%	134	3.84
University of Illinois Urbana-Champaign	606	241	39.77%	167	3.82
New York University	780	316	40.51%	174	3.76
University of Toronto	901	377	41.84%	249	3.76
Columbia University	962	433	45.01%	294	3.76
University of Oxford	589	307	52.12%	147	4.00

SQL

```
SELECT university,
       entries,
       acceptances,
       ROUND(100.0 * acceptances / NULLIF(entries, 0), 2) AS acceptance_percent,
       accepted_with_gpa,
       ROUND(avg_gpa_accepted::numeric, 2) AS avg_gpa_accepted
FROM (
  SELECT llm_generated_university AS university,
         COUNT(*) AS entries,
```

```

        COUNT(*) FILTER (WHERE status ILIKE 'accept%') AS acceptances,
        COUNT(gpa) FILTER (WHERE status ILIKE 'accept%') AS accepted_with_gpa,
        AVG(gpa) FILTER (WHERE status ILIKE 'accept%') AS avg_gpa_accepted
FROM applicants
WHERE llm_generated_university IS NOT NULL
GROUP BY llm_generated_university
ORDER BY entries DESC
LIMIT 20
) AS most_applied_to
ORDER BY acceptance_percent ASC;

```

WHAT IT DOES

The inner query groups every entry by standardized university and keeps the twenty with the most submissions, so each rate rests on a real sample rather than on a school with three entries. For each it computes the acceptance rate and, separately, the average GPA of just the accepted applicants — two FILTER clauses over the same scan. The outer query ranks by acceptance rate ascending, so the hardest school comes first. Grouping on the standardized column rather than the raw one matters here: the raw field spells the same school several ways, which would split one university across several groups and push all of them out of the top twenty.

READ BEFORE QUOTING THIS NUMBER

An acceptance rate computed from Grad Cafe is not an admissions rate. It is the share of people who chose to post about one school and reported being admitted, and both halves of that are selected: someone holding an offer has more reason to come back and post than someone rejected everywhere. The ordering here is still informative because every school in the table is subject to the same posting behavior, but the levels are not comparable to a published admissions rate.

Question 11 — my own question

Do international applicants fare differently from American ones — and are they equally willing to say what their GPA was?

RESULT

Acceptance rate: 39.23% American vs 35.39% international — a gap of 3.84 points
 But the sharper gap is in what they disclose: 64.78% of American entries report a GPA against 52.98% of international ones
 Yet the GPAs they do report are near identical — 3.77 American, 3.76 international

Cohort	Entries	Acceptances	Acceptance rate	Report a GPA	Avg GPA
American	25,420	9,973	39.23%	64.78%	3.77
International	23,360	8,267	35.39%	52.98%	3.76

SQL

```

SELECT initcap(trim(us_or_international)) AS cohort,
       COUNT(*) AS entries,
       COUNT(*) FILTER (WHERE status ILIKE 'accept%') AS acceptances,
       ROUND(100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%')
             / NULLIF(COUNT(*), 0), 2) AS acceptance_percent,
       ROUND(100.0 * COUNT(gpa) / NULLIF(COUNT(*), 0), 2) AS gpa_disclosure_percent,
       ROUND(AVG(gpa)::numeric, 2) AS avg_gpa
FROM applicants
WHERE us_or_international IS NOT NULL
      AND trim(us_or_international) <> ''
GROUP BY 1
ORDER BY acceptance_percent DESC;

```

WHAT IT DOES

One pass grouped by nationality, computing three things per cohort: the acceptance rate, the share of entries that disclose a GPA at all, and the average of the GPAs that are disclosed. Entries with no nationality are excluded, as in Question 2, so 'did not say' is not silently counted as domestic. The disclosure percentage is the reason the question is worth asking: an average can only describe the people who answered, so knowing how many declined to answer is part of knowing what the average means.

READ BEFORE QUOTING THIS NUMBER

The disclosure gap has an innocent explanation that the acceptance gap does not. Grad Cafe offers a single GPA box with no scale, and much of the world does not grade on a 4.0 — Module 2 found GPAs of 8.25 and 9.10 in this data, which are 10-point CGPAs. An applicant whose GPA does not fit the box has a good reason to leave it empty, so the lower international disclosure rate is at least partly a measurement artefact rather than reticence. That cuts both ways: it also means the international GPA average is drawn from a self-selected subset of international applicants, which is exactly the kind of hidden filtering limitations.pdf is about.